INCH-POUND

MIL-I-24768/12
25 February 1992

SUPERSEDING
MIL-P-79C
15 June 1961

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, PAPER-BASE, PHENOLIC-RESIN (PBM)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a paper cellulosic substrate impregnated and bonded with a phenolic-resin matrix or binder and processed to conform to this specification.

Intended use: The laminated material covered by this specification is primarily intended for mechanical applications where electrical properties are of secondary importance or for electrical applications where no moisture is present. Special precautions should be taken if the material is to be tapped or machined parallel to the laminations to prevent cracking.

Forms: Sheets and rolled tubes are recommended forms of laminated material covered by this specification.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 130, and an electrical strength test index of 130.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/12-S-1	0.010	0.002
M24768/12-S-2	0.015	0.0025
M24768/12-S-3	0.020	0.003
M24768/12-S-4	0.025	0.0035
M24768/12-S-5	0.031	0.0035
M24768/12-S-6	0.047	0.0045
M24768/12-S-7	0.062	0.005
M24768/12-S-8	0.094	0.007
M24768/12-S-9	0.125	0.008
M24768/12-S-10	0.156	0.009
M24768/12-S-11	0.188	0.010
M24768/12-S-12	0.219	0.011
M24768/12-S-13	0.250	0.012
M24768/12-S-14	0.312	0.014
M24768/12-S-15	0.375	0.017
M24768/12-S-16	0.438	0.019
M24768/12-S-17	0.500	0.021
M24768/12-S-18	0.625	0.024
M24768/12-S-19	0.750	0.027
M24768/12-S-20	0.875	0.030
M24768/12-S-21	1.000	0.033
M24768/12-S-22	1.125	0.035
M24768/12-S-23	1.250	0.037
M24768/12-S-24	1.375	0.039
M24768/12-S-25	1.500	0.041
M24768/12-S-26	1.625	0.043
M24768/12-S-27	1.750	0.045
M24768/12-S-28	1.875	0.047
M24768/12-S-29	2.000	0.049
M24768/12-S-30	2.125	0.052
M24768/12-S-31	2.500	0.056
M24768/12-S-32	2.750	0.061
M24768/12-S-33	3.000	0.065
M24768/12-S-34	3.500	0.073
M24768/12-S-35	4.000	0.082
M24768/12-S-36	4.500	0.090
M24768/12-S-37	5.000	0.095
M24768/12-S-38	5.500	0.105
M24768/12-S-39	6.000	0.115
M24768/12-S-40	6.500	0.120
M24768/12-S-41	7.000	0.130
M24768/12-S-42	7.500	0.140
M24768/12-S-43	8.000	0.145
M24768/12-S-44	8.500	0.155
M24768/12-S-45	9.000	0.160
M24768/12-S-46	9.500	0.170
M24768/12-S-47	10.000	0.180
M24768/12-S-48	_[1]	_[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/12-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Diameter of rolled tubes: The range of sizes for rolled tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled tube with an inside diameter of 0.125 inch and an outside diameter of 0.145 inch is: M24768/12-TR-0.125/0.145. The wall thickness tolerances for tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	48.000	0.145	49.500	0.010	0.750	1/4

TABLE III. *Rolled tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)
0.010 to 0.015	0.003
0.016 to 0.030	0.005
0.031 to 0.061	0.006
0.062 to 0.124	0.007
0.125 to 0.249	0.009
0.250 to 0.500	0.011

Performance requirements: Performance requirements for laminated materials shall be as specified in tables IV and V.

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The following performance requirements do not apply to this plastic laminate:

a. Dielectric breakdown
b. Surface resistance
c. Volume resistivity
d. Arc resistance
e. Flammability
f. Silicone content
g. Thermal endurance
h. Tracking resistance
i. Dissipation factor
j. Dielectric constant

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE IV. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage	A D-48/50	4.6.4	Min. kV										
Dielectric constant	A D-24/23	4.6.6	Max.										
Dissipation factor	A D-24/23	4.6.6	Max.										
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2]													
Lengthwise:	E-48/50	4.6.11	Min. ft-lbs. per inch				0.55	0.55	0.55	0.55	0.55	0.55	0.55
Crosswise:							0.50	0.50	0.50	0.50	0.50	0.50	0.50
Flexural strength[3]													
Lengthwise:	A	4.6.12	Min. psi	22,000	25,000	25,000	25,000	25,000	25,000	24,000	24,000	22,000	22,000
Crosswise:				20,000	22,000	22,000	22,000	22,000	22,000	21,000	21,000	19,000	19,000

See footnotes at end of table.

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TABLE IV. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength[3]	A D-48/50	4.6.15	Min. pounds							700 400	700 400	700 400	700 400
Water absorption	D_1-24/23	4.6.17	Max. percent	8.00	6.00	4.20	3.30	2.30	1.80	1.10	0.85	0.75	0.75
Silicone content	E-168/185	4.6.18	Min. sec.										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Impact strength values are not applicable to thickness under 0.251 inch.
[3]Maximum thickness tested shall be 2.000 inches.

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TABLE V. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)				
					0.031–0.047	0.048–0.062	0.063–0.094	0.095–0.125	0.126–0.188
Dielectric strength[1]	A	4.6.5	Min. volts per mil	0.125–8.000	400	400	325	325	200
Compressive strength[2]	A	4.6.14	Min. psi	0.125–0.499	10,000	10,000	10,000	10,000	10,000
				0.500–8.000	10,000	10,000	12,000	12,000	12,000
Specific gravity	A	4.6.16	Min.	0.125–0.499	1.10	1.10	1.10	1.10	1.10
				0.500–8.000	1.12	1.12	1.12	1.12	1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125–0.499	8.0	8.0	7.0	6.0	5.2
				0.500–8.000	8.0	8.0	5.0	4.3	4.0

TABLE V. *Performance requirements for rolled tubes – Continued.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)			
					0.189–0.250	0.251–0.375	0.376–0.500	0.501–0.750
Dielectric strength	A	4.6.5	Min. volts per mil	0.125–8.000	200	145	145	120
Compressive strength[2]	A	4.6.14	Min. psi	0.125–0.499	10,000	10,000	10,000	10,000
				0.500–8.000	12,000	12,000	12,000	12,000
Specific gravity	A	4.6.16	Min.	0.125–0.499	1.10	1.10	1.10	1.10
				0.500–8.000	1.12	1.12	1.12	1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125–0.499				
				0.500–8.000	3.5	3.0	2.4	2.0

[1]The dielectric strength decreases markedly under humid conditions.
[2]Values apply to inside diameters of 0.250 inch and over and to outside diameters of 2.000 inches and less.

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Air Force – 20

Review Activities:
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User Activities:
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Navy – MC, AS

Preparing Activity
Navy – SH
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